Decision and Risk Analysis

Lecture 2: **Bayes' Rule**

DAEN 427/ISEN 427/ISEN 627

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```r
# Load packages
library(bayesrules)
library(tidyverse)
library(janitor)

# Import article data
data(fake_news)

fake_news %>%
  tabyl(type) %>%
  adorn_totals("row")
#  type   n percent
#  fake  60     0.4
#  real  90     0.6
# Total 150     1.0
```

Question 4

Let's suppose that you are an air traveler along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage.

A recording piped through the speakers reminds you that: *"Many bags look alike. Please check your bag carefully before exiting the terminal."*

Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case used by 5 percent of all travelers.

Given your distance from the luggage chute, you can discern only the shape, size, and color of the bags that emerge, not any detailed markings on the bags.

[5a.] Let's suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

[5b.] Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the 99th bag, which also matches ours in shape, size, and color, is our own.

[a.] Let's suppose that the first bag from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).

Answer: Let Y = first bag is yours, and M = first bag matches.

$$P(Y) = \frac{1}{100}$$
$$\mathcal{L}(Y^c \mid M) = P(M \mid Y^c) = 0.05$$
$$\mathcal{L}(Y \mid M) = P(M \mid Y) = 1$$

$$P(Y \mid M) = \frac{P(M \mid Y)P(Y)}{P(M)} = \frac{P(M \mid Y)P(Y)}{P(M \mid Y)P(Y) + P(M \mid Y^c)P(Y^c)} =$$
$$= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168$$

So even though the first bag matches, it is probably not yours.

[b.] Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the 99th bag, which also matches ours in shape, size, and color, is our own.

Answer:

$$P(Y \mid M) = \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining}-1}{\text{total of bags remaining}}}$$

(if 2 bags remain)

$$= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} = \frac{1}{1 + 0.05} \approx 0.9524$$

There is a 95.24% chance that the 99th bag is mine given that it looks like mine.